Ashutosh Lal

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EDUCATION

Aalto University Aug 2023 - Present
Bachelor of Science
Major: Quantum Technology
Minor: Computer Science

Relevant Courses
Machine Learning, Deep Learning, Software Engineering, Software Engineering with LLMs, Operating Systems, Information Security, Python for Engineers, Quantum Mechanics, Quantum Information

EXPERIENCE

Bachelor's Thesis Researcher – Quantum Machine Learning Jun-Oct 2025
- Investigated conjectured quantum advantage in quantum kernel methods.
- Built a custom coherent noise model from scratch in Qiskit to evaluate NISQ-era robustness.
- Compared QSVM performance against classical baselines under realistic hardware constraints.

Aaltoes Spark Participant (2-week MVP Sprint Program) Mar 2026
- Built and shipped a working AI prototype under a two-week deadline with peer feedback and structured workshops.
- Rapidly learned and applied HPC/SLURM tooling while developing an end-to-end fine-tuning pipeline.
- Worked in a cohort-based build environment focused on fast iteration, feedback, and user-centered product thinking.

PROJECTS (*more info on github*)

Agentic RAG Research Assistant| *Langraph, RAG, Multiagent systems* April 2026
- Built a research assistant for ~500 quantum ML papers using deterministic graph retrieval and bounded multi-agent reasoning.
- Improved reliability through explicit uncertainty handling, hybrid dense+sparse retrieval, and state-aware orchestration.

HPC-based fine-tuning pipeline| *LoRA, SLURM, Gemma, Qwen* Mar 2026
- Designed a continual-learning pipeline enabling a 12B LLM to evolve with a growing research corpus.
- Built a full HPC training workflow on CSC Puhti using synthetic data generation, semantic sampling, and parameter-efficient fine-tuning.

ChewMonitor | *neural networks, computer vision, facial recognition, TCNs, mediapipe* Dec 2025
- Built a real-time chewing detection tool to improve eating habits through behavioral feedback.
- Designed a user-facing ML application using facial detection and temporal neural networks.

SKILLS

AI & Prototyping: LLMs, RAG, LangGraph, Prompt Engineering
Programming: Python, Scala, HTML, C, MATLAB
Tools: Linux, Git, SLURM, Jupyter, VS Code
Machine Learning: PyTorch, TensorFlow, scikit-learn, FAISS, Qiskit

ACTIVITIES

Certified Yoga Instructor (RYT 200)
• Experience teaching and communicating complex ideas to diverse groups.
Vipassana Meditation Practitioner
• Developed strong focus, discipline, and long-form independent work habits.